MIL-A-60147A(MU)
11 July 1966
SUPERSEDING
MIL-A-60147(MU)
6 January 1965

MILITARY SPECIFICATION

AMPLIFIER: 10527248

1. SCOPE

1.1 This specification covers one type of amplifier used in the Antioscillation Sighting System XM58 employed in a UH-1B Helicopter. This unit is the integrating and voltage amplifying segment of the amplifier which energizes the azimuth and elevation control motors (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on the date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-T-152	Treatment, Moisture-and Fungus-Resistant, of Communication, Electronic and Associated Electrical Equipment
MIL-I-6181	Interference Control Requirements, Aircraft Equipment
MIL-F-13926	Fire Control Materiel, General Specification Governing the Manufacture and Inspection of
MIL-P-14232	Parts, Equipment and Tools for Ordnance Materiel, Packaging of
MIL-I-45607	Inspection Equipment, Supply and Maintenance of

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables For Inspection by Attributes
MIL-STD-109	Quality Assurance Terms and Definitions

DRAWINGS

U.S. Army, Frankford Arsenal

F10527248	Amplifier 10527248

FSC 1270

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PACKAGING DATA SHEET

MIL-P-14232/P10527248 Packaging of Amplifier: 10527248

(Copies of specifications, standards, drawings, and packaging data sheets required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 First article testing.- Requirements for the submission of first article samples by the contractor for testing shall be as specified in the contract.

3.2 Fabrication.- The amplifier shall be fabricated in accordance with Drawing F10527248 and drawings pertaining thereto.

3.2.1 Treatment, moisture and fungus-resistant.- The amplifier shall be treated for moisture and fungus resistance in accordance with MIL-T-152.

3.3 General specification.- The contractor shall be responsible for adherence to the following requirements of MIL-F-13926:

a. Dimensions and tolerances
b. Inorganic protective surface finishes
c. Part identification and marking
d. Workmanship
e. Soldering electrical connections

3.4 Environmental.

3.4.1 Storage temperatures.- The amplifier shall meet the performance requirements of 3.5 at 60° to 90° Fahrenheit (F) after being subjected to ambient temperatures at -80° and at +160°F.

3.4.2 Operating temperatures.- The amplifier shall meet the performance requirements of 3.5 while exposed to temperatures at -40° and at +150°F.

3.4.3 Altitude.- The amplifier shall meet the performance requirements of 3.5 at an altitude of 30,000 feet (ft) above sea level.

3.4.4 Shock.- The amplifier shall withstand impact shocks up to 15 G's; each shock impulse having a time duration of 11 ± 1 milliseconds (ms). Subsequently, the amplifier shall meet the performance requirements of 3.5.

3.4.5 Vibration.- The amplifier shall meet the performance requirements of 3.5 after being subjected to vibrations up to 5 G's over a requency range of 5 to 500 cycles per second (cps).

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3.5 Performance.

3.5.1 Feedback signal gain.- The gain between the feedback signal input and the amplifier output shall be equal to +1.1 $\frac{VDC}{VDC}$ + 15% for outputs from +2 to -2 Vdc.

3.5.2 Feedback signal input sensing.- The amplifier output shall be a positive voltage with respect to signal ground for a positive voltage at the input.

3.5.3 Gimbal position control signal gain.- The gain between the a-c gimbal position control signal input and the amplifier output shall be equal to 7.5 $\frac{VDC}{VAC}$ ± 20% for outputs from +2 to -2 Vdc.

3.5.4 Trim signal gain.- The yaw between the a-c trim signal input and the amplifier output shall be equal to 2.4 $\frac{VDC}{VAC}$ ± 15% for outputs from +2 to -2 Vdc.

3.5.5 Gyro pick-off signal gain.- The gain between the a-c gyro pick-off signal input and the amplifier output shall be equal to 22.5 $\frac{VDC}{VAC}$ ± 10% for outputs from +2 to -2 Vdc.

3.5.6 A-C signal input sensing.- The amplifier output voltage shall be negative with respect to signal ground when each a-c input signal is in phase with the square-wave reference voltage.

3.5.7 Offset.- With all four input terminals grounded, the output shall be equal to or less than ± 30 millivolts (mv) dc.

3.5.8 Linearity.- Linearity shall be ± 10% for an output voltage range of ± 2.0 Vdc for each input channel.

3.5.9 Saturation.- The output voltage shall be greater than 2.5 Vdc for gyro pick-off signal inputs of 4 Vac or more.

3.5.10 Caging input.- The gain between the caging input and the amplifier output shall be equal to 20 ± 20% volts DC/volt AC for an output of approximately ± 2 volts DC.

3.6 Radio interference suppression.- Interference control requirements shall be in accordance with MIL-I-6181.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inpsections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.1.1 General provisions.- The component and subassembly inspection requirements of MIL-F-13926 form a part of the Quality Assurance Provisions of this specification. Definitions of inspection terms shall be as listed in MIL-STD-109.

4.2 First article testing.- When the contract specifies testing of preproduction or initial production samples, the following paragraphs shall apply as applicable.

4.2.1 Preproduction test.- Preproduction test samples shall be submitted in accordance with contract requirements. Unless otherwise specified in the contract, two preproduction samples shall be submitted to the procuring agency for approval of design and construction features. Preproduction samples shall meet all inspection provisions of 4.3, 4.4 and 4.6 as well as any additional tests that may be specified in the contract.

4.2.2 Initial production test.- Initial production test samples shall be submitted in accordance with contract requirements. Unless otherwise specified in the contract, samples shall be the first two production units and shall have been manufactured using the same materials, processes and procedures as will be used in regular production. Initial production testing shall be in accordance with 4.3, 4.4 and 4.6 and shall be witnessed by the Government representative for allexaminations and tests. The Government reserves the right to conduct any or all examinations or tests as may be deemed necessary.

4.2.3 First article disapproval.- Failure of any one of a first article sample submitted shall be cause for disapproval by the Government and subsequent action as required by the contract clauses applicable to first article testing.

4.3 Inspection provisions.

4.3.1 Submission of product.- Unless otherwise specified by the contracting officer, inspection lot size, lot formation and presentation of lots shall be in accordance with "Submission of Product" provisions of MIL-STD-105.

4.3.2 Examination and tests.

4.3.2.1 Components and subassemblies.- All components and subassemblies shall be inspected in accordance with the inspection provisions contained in Supplementary Quality Assurance Provisions (SQAP) listed in the List of Inspection Documents (IEL), when provided with the technical data package (TDP). In the absence of SQAP's the applicable Quality Assurance Provisions of MIL-F-13926 shall apply.

4.3.2.2 First article tests.- Unless otherwise specified, the requirements and tests of Table I shall be conducted on first article tests only. In the event that the processes, materials or other changes in design or manufacture occur subsequent to first article approval, the first items so produced shall be submitted to the Government in accordance with 4.2 for approval prior to resumption of production.

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TABLE I

	CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
301.	Altitude	3.4.3	4.6.2.3
302.	Shock	3.4.4	4.6.2.4
303.	Vibration	3.4.5	4.6.2.5
304.	Radio interference suppression	3.6	4.6.4

4.3.2.3 Final acceptance inspection.- Subsequent to first article approval, examination and tests related to Section 3 herein shall be performed on a single defect (individual characteristic) basis in accordance with MIL-STD-105 and the sampling plans specified in Tables II and III herein. Examination and tests for Packaging, Packing and Marking shall be in accordance with MIL-P-14232 and Section 5 herein. The tabulated classification of defects in Tables II and III shall constitute the minimum inspection to be performed by the supplier after first article approval and prior to Government acceptance or rejection by item or lot.

TABLE II - CLASSIFICATION OF DEFECTS

CLASS		CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
CRITICAL:		NONE		
MAJOR:		AQL 1.5% Defective		
	101.	Moisture and fungus resistance	3.2.1	4.6.1
	102.	Feedback signal gain	3.5.1	4.6.3.1
	103.	Feedback signal input sensing	3.5.2	4.6.3.1
	104.	Gimbal position control signal gain	3.5.3	4.6.3.2
	105.	Trim signal gain	3.5.4	4.6.3.3
	106.	Gyro pick-off signal gain	3.5.5	4.6.3.4
	107.	A-C signal input sensing	3.5.6	4.6.3.2, 4.6.3.3, 4.6.3.4
	108.	Offset	3.5.7	4.6.3.5
	109.	Linearity	3.5.8	4.6.3.6
	110.	Saturation	3.5.9	4.6.3.7
	111.	Caging input	3.5.10	4.6.3.8
MINOR:		NONE		

4.3.3 Acceptance and rejection.- Rejected lots shall be screened for all defective characteristics. Removal or correction of defective units and resubmittance of rejected lots shall be in accordance with "Acceptance and Rejection" as specified in MIL-STD-105.

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4.4 Special sampling.- Three amplifiers shall be selected at random as a special sampling from each 100 produced. The sample shall meet the requirements and tests in Table III.

TABLE III

	CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
401.	Design and construction	3.2	Applicable drawing - visual
402.	General specification	3.3	MIL-F-13926 - visual
403.	Temperature (storage)	3.4.1	4.6.2.1
404.	Temperature (operating)	3.4.2	4.6.2.2

4.4.1 Failure of sample.- Should any one item of a special sampling fail to meet the specified test requirements, acceptance of the product shall be suspended by the Government until necessary corrections have been made by the contractor and resubmitted samples have been approved (see 4.3.3.).

4.5 Inspection equipment.- Except as otherwise provided for by the contract, the contractor shall supply and maintain inspection equipment in accordance with the applicable requirements of MIL-I-45607.

4.5.1 Government furnished inspection equipment.- Where the contract provides for Government furnished test equipment, supply and maintenance of test equipment shall be in accordance with the applicable requirements specified in MIL-I-45607.

4.5.2 Contractor furnished inspection equipment.

4.5.2.1 Government design.- Unless otherwise specified in the contract, all inspection equipment specified by drawing number in specifications or SQAP forming a part of the contract shall be supplied by the contractor in accordance with technical data listed in the List of Inspection Documents when provided with the technical data package (TDP).

4.5.2.2 Contractor design.- The contractor shall design and supply inspection equipment compatible with the "Test Methods and Procedures" specified in 4.6 of this specification and with the component inspection procedures specified in "Examination" and "Test Facilities" requirements of MIL-F-13926. Since tolerance of test equipment is normally considered to be within 10% of the product tolerance for which it is intended, this inherent error in the test equipment design must be considered as part of the prescribed product tolerance limit. Thus, concept, construction, materials, dimensions and tolerances used in the design of test equipment shall be so selected and controlled as to insure that the test equipment will reliably indicate acceptability of a product which does not exceed 90% of the prescribed tolerance limit, and permit positive rejection when non-conforming. Construction shall be such as to facilitate routine calibration of test equipment.

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4.6 Test methods and procedures.

4.6.1 Treatment, moisture and fungus-resistant.- The amplifier shall be tested in accordance with MIL-T-152 to insure compliance with 3.2.1.

4.6.2 Environmental.

4.6.2.1 Storage temperatures.- The amplifier, while in a non-operating condition, shall be subjected to the Temperature Cycling Test, Procedure II of MIL-F-13926 using Conditions A and H. Upon stabilization at ambient room temperature, the amplifier shall meet the performance requirements specified in 3.5.

4.6.2.2 Operating temperatures.- The amplifier shall be subjected to the Temperature Cycling Test, Procedure 1 of MIL-F-13926 using Conditions C and G. During temperature stabilization at Conditions C, G and room ambient, the amplifier shall meet the performance requirements specified in 3.5.

4.6.2.3 Altitude test.- The amplifier shall be placed within a test chamber and the absolute internal pressure reduced to 8.88 inches of mercury (corresponding to an altitude of 30,000 feet above sea level). After sufficient time has elapsed to allow all entrapped air to escape, the amplifier shall meet the requirements specified in 3.5.

4.6.2.4 Shock test.- The amplifier shall be tested in accordance with MIL-F-13926, Procedure II. The amplifier shall not suffer damage or subsequently fail to provide the performance specified in 3.5.

4.6.2.5 Vibration test.- With the amplifier mounted and operating at room ambient conditions, the able shall be vibrated along a horizontal direction with the frequency varying between 5 and 500 cps. For frequencies between 5 and 20 cps, the double amplitude shall be 0.1 inch. For frequencies between 20 and 33 cps, the amplitude shall be varied so as to maintain a 2 G acceleration. For frequencies between 33 and 52 cps, the double amplitude shall be 0.036 inch. For frequencies between 52 and 500 cps, the amplitude shall be varied so as to maintain a 5 G acceleration. The rate of frequency change shall be logarithmic and shall be such that a complete cycle (5-500-5) will consume approximately 15 minutes. The test shall continue for four 15-minute cycles for a total of 1 hour. (When provisions for logarithmic cycling is not available, a linear rate of frequency change may be used.) The cycling test shall be repeated with the axis of vibration changed 90 degrees in the horizontal direction and then with the axis of vibration changed to vertical. At the completion of each direction of vibration, the amplifier shall be closely inspected for any evidence of mechanical failure, loose or broken parts. At end of test, the amplifier shall meet the performance requirements of 3.5.

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4.6.3 Performance.

4.6.3.1 Feedback signal gain and sensing test.- Use test set up shown in Figure 1. Apply maximum input voltages of + and -1.5 Vdc to the feedback input terminals (pins R and P). Measure the d-c voltage across the output (pins X and Y) for each polarity of input. Calculate the gain (output Vdc/input Vdc) for compliance with 3.5.1 and observe the feedback signal input sensing for compliance with 3.5.2.

4.6.3.2 Gimbal position control gain and sensing test.- Use test set up shown in Figure 1. Apply a 0.06 volt root-mean square (rms) maximum input signal of 0° and 180° phase with respect to the 400 cps reference to the gimbal position control signal terminals (pins A and H). Measure the d-c voltage across the output (pins X and Y) for each phase of input signal. Calculate the gain (output Vdc/input Vac) for compliance with 3.5.3 and observe the a-c signal input sensing for compliance with 3.5.6.

4.6.3.3 Trim signal gain and sensing test.- Use test set up shown in Figure 1. Apply a 0.50 volt rms maximum input signal of 0° phase and 180° phase with respect to the 400 cps reference to the trim signal terminals (pins C and H). Measure the d-c voltage across the output (pins Y and X) for each phase of input signal. Calculate the gain (output Vdc/input Vac) for compliance with 3.5.4 and observe the a-c signal input sensing for compliance with 3.5.6.

4.6.3.4 Gyro pick-off signal gain test.- Use test set up shown in Figure 1. Apply a 0.08 volt rms maximum input signal of 0° phase and 180° phase with respect to the 400 cps reference to the gyro pick-off signal terminals (pins E and H). Measure the d-c voltage across the output (pins Y and X) for each phase of input signal. Calculate the gain (output Vdc/input Vac) for compliance with 3.5.5 and observe the a-c signal input sensing for compliance with 3.5.6.

4.6.3.5 Offset test.- Use test set up shown in Figure 1. Set switch S1 to Offset Test, then with all four input terminals grounded (pins R, A, C, E, to H). Measure the d-c voltage across the output (pins Y and X) to insure compliance with 3.5.7.

4.6.3.6 Linearity test.- Use test set up shown in Figure 1. Plot a gain curve for each of the four inputs for output voltages up to ± 2.0 Vdc. Establish the best straight line for each to insure compliance with 3.5.8.

4.6.3.7 Saturation test.- Use test set up shown in Figure 1. Apply 4 V rms to the gyro pick-off signal input (pin E). Measure the d-c output voltage (pins Y and X) to insure compliance with 3.5.9.

4.6.3.8 Caging input.- Use test set up shown in Figure 1. Apply a 2 volt rms maximum input signal 0° and 180° phase with respect to the 400 cps referenced to synchro caging terminal (pins B and H). Measure the d-c voltage across output terminals X and Y for each phase of input. Calculate the gain output (VDC/input VAC) for compliance with 3.5.10 and observe the a-c signal input sensing for compliance with 3.5.6.

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4.6.4 Radio interference test.- The amplifier shall be tested in accordance with MIL-I-6181 to insure conformance with 3.6.

5. PREPARATION FOR DELIVERY

5.1 Packaging, packing and marking.- Packaging, packing and marking shall be in accordance with MIL-P-14232 and Packaging Data Sheet MIL-P-14232/P10527248. The level of protection shall be as specified in the procurement document.

6. NOTES

6.1 Intended use.- This Amplifier 10527248 is used in the Electronics Control Amplifier 10527247 of the Antioscillation Sighting System XM58. It receives the rate gyro pick-off signal along with the gimbal position signal, a drift compensating trim signal, and a stabilizing feedback signal originating from the control motor voltage. The amplifier modulates the latter signal, sums it with the other three signals, and amplifies the resultant signal. The amplified signal is then demodulated, filtered and fed as an input to Amplifier 10527249.

6.2 Ordering data.- Procurement documents should specify the following:

a. Title, number and date of this specification
b. Selection of applicable levels of preservation, packaging, and packing (see 5.1).
c. Applicable packaging data sheet number (see 5.1).
d. Applicable stock number

Custodian:

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Preparing activity:

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Project No. 1270-A031Y

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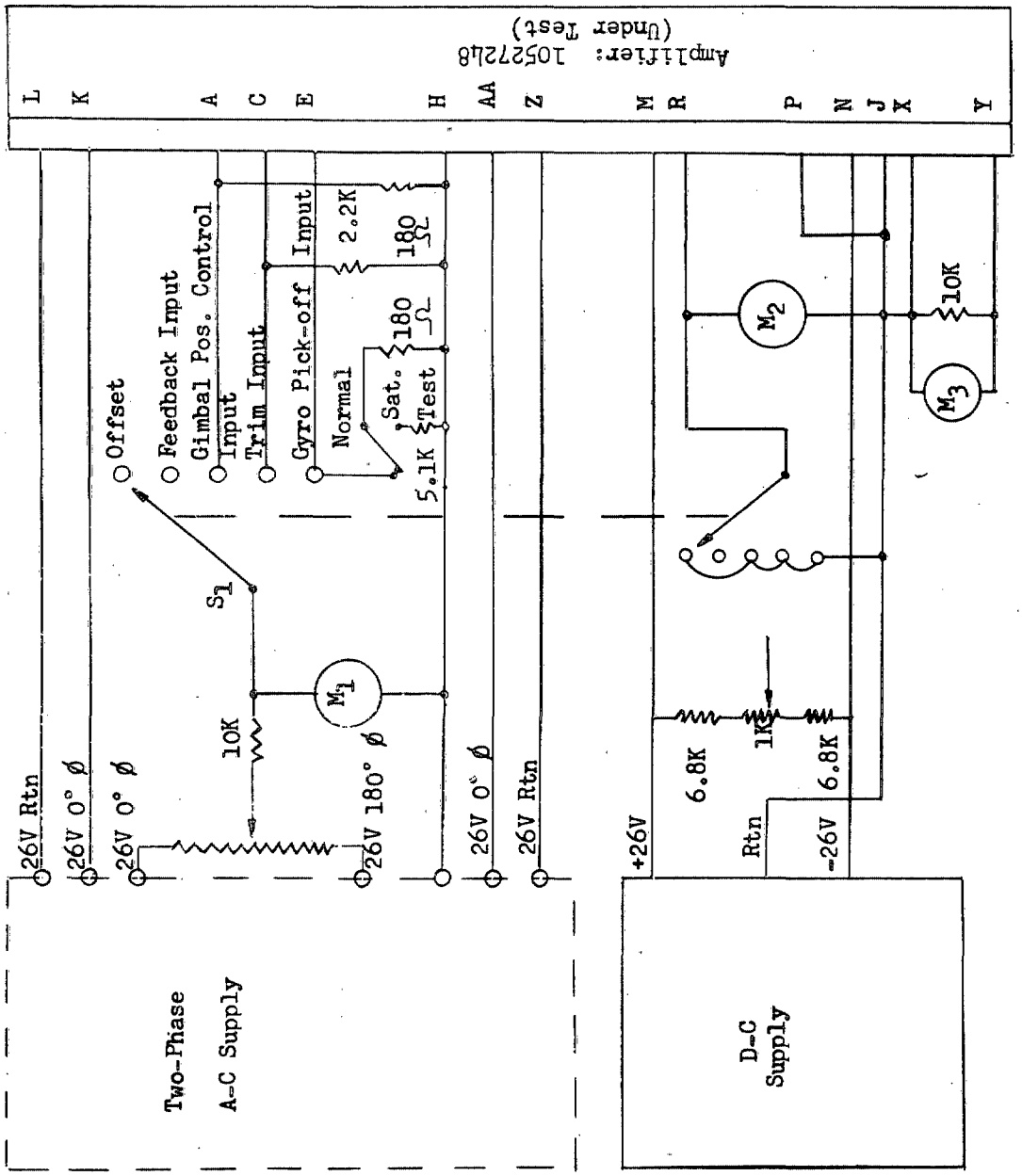


Figure 1. Test Setup for Amplifier: 10527248

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity.

SPECIFICATION

ORGANIZATION | CITY AND STATE

CONTRACT NO. | QUANTITY OF ITEMS PROCURED | DOLLAR AMOUNT

$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)* | DATE

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